

Part B

87

* Sufficiency & Completeness:-

$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(1, \beta)$

$\Rightarrow$ Exponential distribution $\text{Exp}(1/\beta)$ with density:

$$f(x; \beta) = \frac{1}{\beta} \exp\left(-\frac{1}{\beta}x\right); \quad x > 0.$$

$\Rightarrow$ 1 parameter exponential density with $C(\beta) = -\frac{1}{\beta}$ and

$d(x) = x$. $\therefore$ The complete & sufficient statistic for β is

$$T = \sum_{i=1}^n X_i$$

The distribution of T is sum of n iid $\text{Exp}(1/\beta)$ variables

$\Rightarrow$ $\text{Gamma}(n, \beta)$ distribution.

* Unbiased Estimator for β^2

We need to find $g(T)$ such that $E[g(T)] = \beta^2$

$\rightarrow$ Since T is the complete & sufficient statistic, UMVUE $\Rightarrow \hat{h}(\beta) = g(T)$

Let X_1 & X_2 be the first 2 observations of X .

$$\begin{aligned} \therefore E[X_1] &= \beta \text{ (mean of } \text{Exp}(1/\beta)) \\ E[X_1^2] &= \text{Var}(X_1) + (E[X_1])^2 = \beta^2 + \beta^2 = 2\beta^2 \end{aligned} \quad \left[\begin{array}{l} \because \text{Var}(X_i) = \beta^2 \text{ for} \\ \text{Exp}(1/\beta) \end{array} \right]$$

Consider the estimator $W = X_1 X_2$; [X_1 & X_2 are independent]

$$\Rightarrow E[W] = E[X_1 X_2] = E[X_1] E[X_2] = \beta \cdot \beta = \beta^2$$

$\therefore W = X_1 X_2$ is an unbiased estimator of β^2

* Finding UMVUE By Lehmann-Schutte' (Rao-Blackwellization)

by conditioning unbiased estimator $W = X_1 X_2$ on complete & sufficient statistic T :

$$\hat{h}(\beta) = E[W|T] = E[X_1 X_2 | T]$$

Since T is complete & sufficient statistic this conditional Expectation does not depend on β & is the UMVUE.

P.T.O.

Q7 p2.

Using symmetry property (all X_i are exchangeable)

$$E[X_i X_j | T] = E[X_1 X_2 | T] \text{ for } i \neq j$$

Consider T^2 :

$$T^2 = \left(\sum_{i=1}^n X_i \right)^2 = \sum_{i=1}^n X_i^2 + \sum_{i \neq j} X_i X_j$$

$$\therefore \sum_{i \neq j} X_i X_j = T^2 - \sum_{i=1}^n X_i^2$$

$$\therefore E\left[\sum_{i \neq j} X_i X_j | T\right] = E\left[T^2 - \sum_{i=1}^n X_i^2 | T\right]$$

$$\therefore \sum_{i \neq j} E[X_i X_j | T] = T^2 - E\left[\sum_{i=1}^n X_i^2 | T\right]$$

$\therefore$ there are $n(n-1)$ pairs of distinct $X_i X_j$

$$\Rightarrow n(n-1) E[X_1 X_2 | T] = T^2 - E\left[\sum_{i=1}^n X_i^2 | T\right]$$

$\therefore T$ is complete & sufficient, $E[X_1^2 | T]$ must be UMVUE for $E[X_1^2] = 2\beta^2$

UMVUE for β is $\hat{\beta} = T/n = \bar{X}$.

" for β^2 is $\hat{\beta}^2 = \frac{T^2}{n(n+1)}$

" " $2\beta^2$ is $= \frac{2T^2}{n(n+1)}$

$\therefore$ By the uniqueness of UMVUE.

$$E[X_1^2 | T] = \text{UMVUE}(2\beta^2) = \frac{2T^2}{n(n+1)}$$

07 P2

∴ Substituting this back into the symmetry eqn. & simplifying the sum $\sum_{i=1}^n E[X_i^2 | T]$:

$$\sum_{i=1}^n E[X_i^2 | T] = n \cdot E[X_1^2 | T] = \cancel{n \cdot E[X_1^2 | T]}$$

$$\Rightarrow n(n-1) E[X_1 X_2 | T] = T^2 - n \cdot \frac{2T^2}{n(n+1)} = T^2 - \frac{2T^2}{n+1}$$
$$= T^2 \left(\frac{n+1-2}{n+1} \right) = T^2 \left(\frac{n-1}{n+1} \right)$$

* Solve for UMVUE

$$\hat{h}(\beta) = E[X_1 X_2 | T] = \frac{T^2 \left(\frac{n-1}{n+1} \right)}{n(n-1)} = \frac{T^2}{n(n+1)}$$

Substituting $T = \sum_{i=1}^n X_i$

$$\hat{h}(\beta) = \frac{1}{n(n+1)} \left(\sum_{i=1}^n X_i \right)^2$$